

ASSIGNMENT 1

1. Explain the difference between frontend, backend, and full-stack development with suitable real-world examples.

Ans:- Frontend development is the part that users see and interact with. It includes everything we can click, read, or watch on a screen. For example, when we open a shopping website like Myntra, the colours, buttons, images, and layout we see are made by frontend developers using HTML, CSS, and JavaScript.

Backend development deals with databases, servers and logic of the website, it manages data and makes sure everything works properly. It is not visible to users, for example when we login to Netflix, the backend checks the username and password and then shows the profile and watchlist.

Full stack development means working on both frontend and backend, a full stack developer can design how the website looks and also make it work from the server side. For example, a full stack developer can create a shopping app from designing it to managing the databases and its behind logic.

2. Describe how a browser requests and displays a web page from a web server.

Ans:- When we type a website's address (like www.google.com) into a browser and press Enter, a process starts behind the scenes to bring that page to our screen.

First, the browser sends a request to the web server where that website is stored. This request travels through the internet. The browser basically asks the server, "Please send me the files for this web page."

Then, the web server receives this request, finds the required files (like HTML, CSS, JavaScript, and images), and sends them back to the browser.

After receiving these files, the browser starts to build the web page:

- It reads the HTML to know the structure of the page.

- It uses CSS to add colors, fonts, and layout.
- It runs JavaScript to make the page interactive (like buttons or animations).

3. Identify and list the tools required to set up a web development environment. Explain the purpose of each.

Ans:- list of tools required are:-

1. Code editor- this is the platform where you'll write all your code like of HTML, CSS, JavaScript. For example, Visual studio code, sublime text, etc.
2. Web browser and developer tools- you will need a web browser to view the website that you have built. Modern browsers also come with built-in "Developer Tools" (often called DevTools) that allow you to inspect elements on your page, debug JavaScript, see how your CSS is applied, and check for errors. For example:- Google chrome
3. Terminal- This is a text-based interface where you can type commands to interact with your computer. You'll use it for tasks like running development servers, installing software packages, and managing your project's files.
4. Node.js and npm- node.js allows you to run javascript code outside your web browser. npm is a package manager that comes with Node.js and lets you easily install and manage libraries and tools for your projects.

4. Explain what a web server is and give examples of commonly used servers.

Ans:- A web server is a computer that stores websites and delivers them to users when they are requested through a browser. In simple words, whenever we open a website, the web server is the place where all that website's data is kept.

Some commonly used web servers are Apache, Nginx, Microsoft IIS, and LiteSpeed.

5. Define the roles of a frontend developer, backend developer, and database administrator in a project.

Ans:-A frontend developer is responsible for the part that users see and interact with. They design and build the layout, buttons, menus, and overall look of the website using HTML, CSS, and JavaScript.

A backend developer works behind the scenes. They handle how the website works, manage the logic, and connect it to the database.

A database administrator (DBA) takes care of the data part. They create, manage, and protect the database where all the website information is stored like user accounts, orders, and messages.

6.Explain the difference between static and dynamic websites. Provide an example of each.

Ans:- Static website is a website at which the content stays the same for everyone until and unless the developer edits it.
For example on a company's website the "about us" page is static as it is same for everyone.

Dynamic websites are the websites at which the content is not fixed , it shows different content to different users.for example:-shopping sites like amazon and filpkart show different data to different people according to their interest.

7.Research and list five web browsers. Explain how rendering engines differ between them.

Ans:-list of five web browsers are:-

1. Google chrome
2. Mozilla firefox
3. Opera
4. Microsoft edge
5. Apple safari

Google chrome and mozilla firefox use blink engine.it is fast and works well with modern websites.

Mozilla firefox uses it's own engine named as gecko.

Apple safari uses the webkit rendering engine.

Opera uses the blink engine just like chrome and edge.

8. Create a simple diagram showing how the client-server model works in web architecture.

